Ringraziamenti

Alla mia famiglia, che è sempre stata il punto fermo di ogni mio traguardo.

A mia madre, Romina, che più di chiunque altro mette sempre gli altri prima di sé, spesso tenendosi dentro preoccupazioni e fatiche pur di vedere sereni coloro che ama. È la persona che chiamo quando qualcosa non va, quando ho bisogno di sfogarmi o semplicemente di sentirmi capita. Il suo amore, la sua forza e la sua presenza saranno sempre il mio rifugio sicuro.

A mio padre, Massimo, che mi ha accompagnata in ogni fase della vita, dalle mattine in cui mi teneva per mano andando alle elementari fino alle sfide di oggi. Fin da bambina, quando qualcosa sembrava più grande di me, è sempre riuscito a rassicurarmi e a insegnarmi che le difficoltà fanno parte della vita e che vanno affrontate con coraggio, sacrificio e determinazione. Dietro questa lezione ci sono sempre stati il suo affetto, la sua dolcezza e la certezza di poter contare su di lui in qualsiasi momento.

A mio fratello, Gabriele, che è sempre stato per me una fonte di orgoglio e un esempio da seguire. La sua dedizione, la sua determinazione e il suo percorso mi hanno spesso ispirata a dare il meglio di me stessa.

A mia nonna Irma, che da sempre rappresenta una delle presenze più importanti della mia vita. Fin dall'infanzia è stata il mio punto di riferimento quotidiano, la persona alla quale ho raccontato ogni gioia, ogni dubbio e ogni difficoltà. È una consigliera, un'insegnante, un'amica sincera e una fonte inesauribile di amore.

A mia nonna Rita, per la sua dolcezza e per l'esempio di gentilezza che mi ha trasmesso nel corso degli anni. Con la sua capacità di riunire la famiglia attorno a una tavola ogni domenica, mi ha insegnato il valore della generosità, della condivisione e dell'amore sincero verso gli altri.

A mio nonno Michele, presenza costante e preziosa nella mia vita. Dietro il suo carattere spesso riservato si nascondono una grande sensibilità, una profonda dolcezza e una straordinaria capacità di stimolare la riflessione attraverso i suoi racconti e i suoi consigli. Il suo affetto e la sua vicinanza hanno per me un valore immenso.

Un ringraziamento speciale va a Lorenzo. Abbiamo condiviso questo percorso fin dal primo giorno: gli stessi esami, le stesse preoccupazioni e le stesse soddisfazioni. Abbiamo affrontato insieme difficoltà e traguardi, sostenendoci reciprocamente in ogni fase del cammino. Mi auguro che il prossimo capitolo della nostra vita insieme sia altrettanto bello, entusiasmante e ricco d'amore quanto quello che ci ha portati fin qui.

Infine, ai miei amici e a tutte le persone che hanno condiviso con me questi anni universitari, grazie per il sostegno, l'affetto e i ricordi che porterò sempre con me.

Contents

List of Figures

List of Tables

Chapter 1

Introduction

Modern finance relies on the assumption that financial returns are normally distributed, which allows standard deviation to be interpreted as a meaningful measure of risk. This thesis evaluates whether this assumption provides a realistic description of financial return behaviour. Normality is a cornerstone of the standard financial framework and underlies both Markowitz's (1952) mean–variance portfolio theory and the Black–Scholes (1973) option-pricing model. Despite its academic success and widespread adoption, the validity of normality has long been questioned: Mandelbrot (1963) documented heavy tails in financial returns, Fama (1965) reported similar evidence for stock markets, and Cont (2001) provided supporting findings across a wide range of markets worldwide. Taken together, these studies show that extreme outcomes occur more frequently than the normal distribution predicts and motivate heavy-tailed alternatives such as stable Paretian models.

Building on this literature, the thesis tests the normality hypothesis using 8,817 observations of the S&P 500 Total Return Index from January 1990 to December 2024, and the analysis proceeds along three dimensions. First, formal statistical tests are used to assess normality, and particular attention is paid to the treatment of crisis periods. The thesis recognises that the mechanisms that amplify crises are endogenous to financial markets. Nevertheless, the baseline analysis is conducted on the full sample, and key tests are replicated on a restricted sample to assess the sensitivity of the results to this choice. Second, alternative heavy-tailed distributions are fitted by maximum likelihood. To ensure a fair comparison across distributions with different numbers of parameters, AIC and BIC are used to penalise complexity by imposing a cost for each additional parameter. On this basis, the Student-t distribution is identified as the best empirical model. Third, a GARCH (1,1) model is used to capture time-varying volatility and to examine the distribution of standardised residuals. Finally, the implied volatility surface of S&P 500 options on 21 March 2025 (OptionMetrics Ivy DB via WRDS) provides complementary evidence. The options data display a pattern that is difficult to reconcile with the normality assumption: the 25-delta risk reversal extends from -4.76% to -6.71% across maturities, and the butterfly spread is positive at all tenors.

The motivation for this thesis is both academic and practical. Academically, Mandelbrot's critique remains largely unincorporated into standard models, which often retain normality as a convenient default despite ongoing debate about its validity. Practically, systematic underestimation of tail risk can have serious consequences. Value at Risk, a central measure in the Basel regulatory framework, is typically computed under a normality assumption. As a result, institutions may hold insufficient capital against extreme losses precisely because such events are treated as exceptionally unlikely. This dynamic can contribute to the onset and amplification of crises, including the 2008 financial crisis. In the empirical results, the GARCH-t yields a VaR violation rate of 1.10% (close to the 1% target), whereas the GARCH-Normal produces 2.27%, more than double. The distributional assumption is therefore not merely an academic detail: it materially affects the credibility and stability of financial risk management.

The remainder of this thesis is organised as follows. Chapter 2 develops the theoretical background. Chapter 3 presents the data and the main empirical results on normality testing and distribution fitting. Chapter 4 analyses volatility dynamics using GARCH models. Chapter 5 connects the historical return evidence to the options market through the implied volatility surface. Chapter 6 concludes.

Chapter 2

Theoretical Background

2.1 The Normality Assumption in Modern Finance

Bachelier's (1900) doctoral thesis paved the way for the use of the Gaussian framework in finance. This marked an early instance of modelling stock price movements as Brownian motion with normally distributed increments. Thereafter, normality became instrumental in the formulation of two cornerstones of modern finance: portfolio theory and derivative pricing. Under the normality assumption, Markowitz (1952) showed that mean and variance are sufficient to characterise portfolio returns, providing the theoretical basis for mean–variance optimisation. Building on this, Black and Scholes (1973) derived their option-pricing formula—now an industry standard—under the premise that log-returns are normally distributed with constant volatility, which delivers a closed-form solution.

Normality has also remained attractive from a modelling perspective because of its mathematical convenience. Properties such as stability, the central limit theorem, and a finite, sample-stable variance make the Gaussian framework tractable and comparatively robust. These considerations offer a rationale for expecting approximate normality in aggregated returns. For a long time, therefore, normality appeared to be a natural default, until the behaviour of the distributional tails became the central focus.

2.2 Mandelbrot's Critique and Heavy-Tailed Alternatives

Normality's empirical inadequacy was documented by Mandelbrot (1963), who analysed daily price changes of cotton in New York from 1900 to 1958. He found that the distribution of returns exhibited much heavier tails than normality allows. As a theoretical alternative, Mandelbrot proposed the class of α-stable distributions, characterised by four parameters: a stability index $\alpha \in (0, 2]$, a skewness parameter $\beta \in [-1, 1]$, a scale parameter σ, and a location parameter μ. Normality belongs to this class as the special case $\alpha = 2$, whereas for $\alpha < 2$ the variance is infinite, raising serious concerns for standard risk measurement. In particular, a risk measure that diverges cannot meaningfully quantify dispersion or support comparisons across assets and periods, which limits its practical usefulness.

Extending this line of evidence, Fama (1965) documented heavy tails and excess kurtosis in equity returns using Dow Jones daily data, providing further support for Mandelbrot's position and empirically confirming Lévy-stable dominance. By contrast, Praetz (1972) proposed the Student-t distribution, with ν degrees of freedom, as a more tractable alternative: it retains finite variance for $\nu > 2$ while still allowing for heavy tails. This makes it possible to accommodate tail risk more realistically while continuing to use variance-based measures of risk. In turn, Bollerslev (1987) was the first to embed the Student-t distribution in a GARCH framework for financial returns, improving its flexibility in applied work.

2.3 The Methodological Question of Crisis Periods

Before turning to the empirical analysis, a recurring concern in the study of return distributions must be addressed: the treatment of extreme observations associated with financial crises. The methodological issue is whether a small number of extreme returns is sufficient to account for non-normality in the data, and whether excluding such observations would restore normality in the remaining sample. One interpretation treats crises as exogenous shocks driven by external factors (e.g., pandemics or geopolitical events) that are unrelated to the return-generating process; under this view, crisis observations should be removed before testing distributional hypotheses. From this perspective, the non-normality of the full sample reflects contamination of an otherwise Gaussian process by outliers.

A competing interpretation holds that crises are not purely exogenous but are amplified by endogenous market mechanisms. Mandelbrot (1963) noted that returns are not independent and can exhibit memory and persistent heavy tails. Cont's (2001) stylised facts similarly emphasise volatility clustering across markets and periods, including tranquil ones. While the initial triggers may be external, falling prices can activate leverage effects and margin calls that force sales independent of fundamentals, which pushes prices down further and induces additional selling. Herding behaviour reinforces this dynamic, as individually rational responses aggregate into market-wide stress. Fire-sale dynamics can complete the spiral, with assets liquidated at distressed prices. Even when the initial shock is exogenous, the amplification mechanism is internal to the financial system. Excluding crisis observations therefore amounts to assuming that these feedback loops are irrelevant to the return-generating process—a strong theoretical restriction that this thesis does not impose.

Nonetheless, results are presented for both the full sample and a restricted sample that excludes the three major crisis windows in the data: the dot-com crash (March 2000–October 2002), the Global Financial Crisis (September 2008–March 2009), and the COVID crash (February–April 2020). This comparison provides a direct check on the robustness of the conclusions to this modelling choice.

2.4 Volatility Clustering and the GARCH Framework

Because financial returns exhibit volatility clustering—"large changes tend to be followed by large changes, of either sign, and small changes tend to be followed by small changes" (Mandelbrot, 1963)—unconditional variance is not, by itself, the most informative risk measure for day-to-day risk management. Conditional variance, by contrast, captures the time-varying nature of financial risk and reflects the prevailing level of volatility given recent market behaviour.

Accordingly, this thesis uses the GARCH framework to examine whether the heavy tails observed in raw returns are a structural feature of the distribution rather than an artefact of omitted time variation in volatility.

In this context, Engle (1982) introduced the Autoregressive Conditional Heteroskedasticity (ARCH) model, in which variance depends on past squared residuals. The original ARCH specification, however, was parameter-intensive and difficult to estimate in practice. Bollerslev (1986) generalised the approach to GARCH (1,1) by adding lagged conditional variance, which parsimoniously captures the persistence of volatility. The resulting conditional variance dynamics can be written as:

$$\sigma_t^2 = \omega + \alpha\varepsilon_{t-1}^2 + \beta\sigma_{t-1}^2 \tag{2.1}$$

where $\varepsilon_t = r_t - \mu$ are mean-adjusted returns, σ_t^2 is the conditional variance, and $\omega > 0$, $\alpha \geq 0$, $\beta \geq 0$, with $\alpha+\beta < 1$, are parameters to be estimated. The parameter α captures the sensitivity of variance to recent shocks, whereas β captures the persistence of the variance process.

2.5 The Implied Volatility Surface and the Volatility Skew

The preceding discussion motivates the limits of normality from the perspective of historical return distributions. Additional, complementary evidence comes from the options market.

In the Black–Scholes framework, for a given underlying and maturity, implied volatility—the value of σ that equates the model price to the observed market price—should be constant across strikes. Empirically, however, when implied volatility is plotted against delta as a measure of moneyness, S&P 500 index options exhibit a pronounced downward slope rather than a flat profile. Out-of-the-money put options therefore trade at higher implied volatilities than at-the-money options or out-of-the-money call options. This pattern, known as the *volatility skew*, is consistent with investors assigning relatively higher probability to large negative outcomes and paying a premium for crash protection. This market-based evidence aligns with the conclusions drawn from the statistical tests.

Two analytical tools are employed in this thesis to interpret the surface: the *risk reversal* ($\text{RR} = \text{IV}_{\text{call}} - \text{IV}_{\text{put}}$ at equal delta), which measures the asymmetry of the smile and is proportional to the skewness of the implied distribution, and the *butterfly spread* ($\text{BF} = (\text{IV}_{\text{call}} + \text{IV}_{\text{put}})/2 - \text{IV}_{\text{ATM}}$), a measure of curvature that is proportional to excess kurtosis.

Chapter 3

Data, Normality Tests and Distribution Fitting

3.1 Data

The dataset comprises 8,817 daily observations of the S&P 500 Daily Total Return Index, sourced from WRDS CRSP, and covers the period from 2 January 1990 to 31 December 2024. The Total Return series accounts for dividend reinvestment and therefore provides a comprehensive measure of equity market performance. Log-returns are defined as $r_t = \ln(P_t/P_{t-1})$, where P_t denotes the index level on day t.

3.2 Descriptive Statistics

Table 3.1 reports summary statistics for the full sample and for a restricted sample that excludes three major crisis windows: the dot-com crash (March 2000–October 2002, $n = 671$), the Global Financial Crisis (September 2008–March 2009, $n = 143$), and the COVID crash (February–April 2020, $n = 42$). The excluded observations total 879, representing 10% of the full sample.

Table 3.1: Descriptive statistics: full sample vs crisis-excluded sample

	Full sample	**Excl. crises**
Observations	8,817	7,938
Mean	0.000466	0.000610
Std. deviation	0.01133	0.00948
Skewness	-0.187	-0.2942
Kurtosis (Pearson)	13.28	6.46
Minimum	-0.1190	-0.0679
Maximum	0.1151	0.0552

Table 3.1 indicates that heavy tails are not solely a crisis-related phenomenon. Crises clearly increase tail mass, but they do not explain it in full. The sample kurtosis falls from 13.28 in the full sample (more than four times the Gaussian benchmark of 3) to 6.46 in the restricted sample, which remains well above the benchmark. This suggests that excess kurtosis persists outside crisis windows and is consistent with the market mechanisms discussed in Section 2.3.

The comparison also shows that skewness is more negative in the crisis-excluded sample (-0.2942) than in the full sample (-0.187). Although this may appear counterintuitive, crisis windows typically include large rebound days as markets partially recover after sharp drawdowns. These rebound observations shift probability mass to the right and partly offset the left tail in the full sample. Once they are excluded alongside the large negative returns, the remaining distribution can appear more left-skewed. This pattern is consistent with the endogeneity view in Section 2.3, in which crisis windows are part of the return-generating process rather than exogenous distortions.

Figure 3.1 plots the time series of daily returns with the three crisis windows highlighted. Large absolute returns are not uniformly distributed over time but are concentrated in relatively short intervals. This volatility clustering has direct implications for modelling: under constant variance, extreme returns would be more evenly dispersed across the sample. Instead, the dot-com crash, the Global Financial Crisis, and the COVID crash feature sustained sequences of large absolute returns separated by extended calm periods. A constant-variance model cannot capture this behaviour, as it would attribute similar variance to, for example, a calm day in July 2017 (when the VIX was below 10) and a day in October 2008 (when the VIX exceeded 80). The GARCH framework, introduced in Section 2.4, addresses this feature by allowing variance to evolve over time as a function of recent shocks and past variance estimates. The evidence in Figure 3.1 therefore provides the empirical motivation for the modelling choice in Chapter 4.

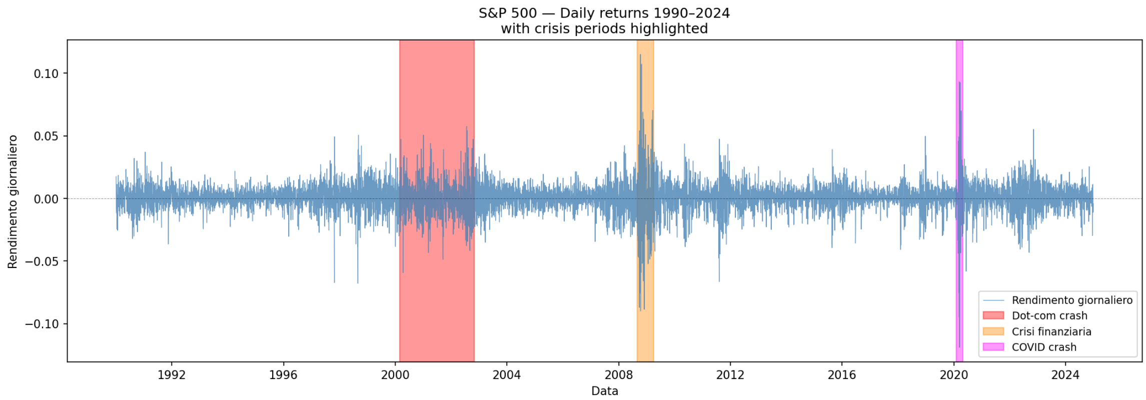

Figure 3.1: S&P 500 daily returns 1990–2024 with crisis windows highlighted. Shaded areas correspond to the dot-com crash (March 2000–October 2002), the Global Financial Crisis (September 2008–March 2009), and the COVID crash (February–April 2020).

Figure 3.2 compares the empirical return distributions across the two samples, with Normal and Student-t fits overlaid. The crisis-excluded sample is visibly less peaked, reflecting the reduction in kurtosis, although both distributions remain leptokurtic relative to the Normal. The comparison therefore reinforces that the rejection of normality is not confined to the full sample.

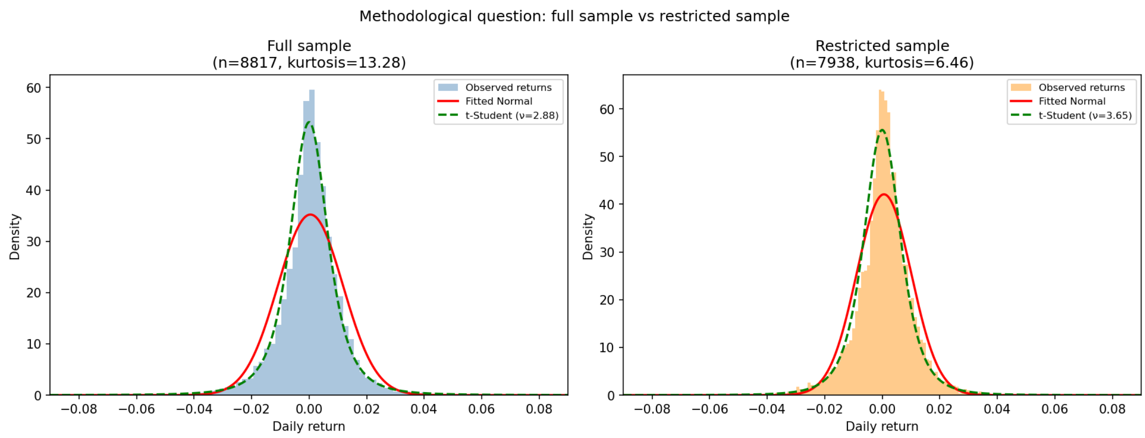

Figure 3.2: Empirical return distributions: full sample (left) and crisis-excluded sample (right), with Normal and Student-t fits overlaid.

3.3 Tests of Normality

3.3.1 Jarque-Bera Test

The Jarque–Bera (1987) test statistic is:

$$\text{JB} = \frac{n}{6}\left(S^2 + \frac{(K-3)^2}{4}\right) \tag{3.1}$$

where S is the sample skewness and K is the sample kurtosis. Under the null hypothesis of normality, JB is asymptotically $\chi^2(2)$.

3.3.2 Anderson-Darling Test

The Anderson–Darling (1952) test statistic is based on the squared difference between the empirical and theoretical cumulative distribution functions, with additional weight assigned to the tails. This feature makes the test well-suited to detecting heavy-tailed departures from normality.

3.3.3 Chi-Square Goodness-of-Fit Test

The chi-square test partitions the support of the distribution into k equiprobable intervals under the null hypothesis and compares observed and expected frequencies. The test statistic follows a $\chi^2(k-1-p)$ distribution, where p is the number of estimated parameters.

Using the three tests jointly is informative because they differ in their sensitivity to specific departures from normality; the combined evidence is therefore more robust than relying on a single test.

3.4 Results

Table 3.2 reports results for both the full sample and the crisis-excluded subsample. In both cases, the null hypothesis of normality is rejected at conventional significance levels. The Anderson–Darling statistic for the full sample (147.96), for example, exceeds the 1% critical value (1.09) by more than two orders of magnitude, indicating that departures from the Gaussian benchmark are concentrated in the tails.

Table 3.2: Normality tests: full sample and crisis-excluded sample

Test	**Full sample** Statistic	p-value	**Excl. crises** Statistic	p-value
Jarque-Bera	38887	< 0.001	4076.3	< 0.001
Anderson-Darling	147.96	< 0.001	71.83	< 0.001
Chi-Square	1184.23	< 0.001	633.56	< 0.001

The Jarque–Bera results likewise reflect substantial excess kurtosis, consistent with heavy tails. The chi-square goodness-of-fit test leads to the same conclusion, with large test statistics and p-values effectively equal to zero.

Although the test statistics decrease when crisis periods are excluded, the rejection of normality remains robust. Non-normality is therefore a persistent feature of the return distribution and cannot be attributed solely to extreme events.

To address the concern that rejections may be driven mechanically by the large sample size, the tests are also repeated on random subsamples of size 1,000 drawn from the full dataset. The null is rejected in virtually all draws, with statistics well above critical values, indicating that the departures from normality are economically meaningful rather than a pure sample-size artefact.

3.5 Fitting Alternative Distributions

The Normal, Student-t, and Lévy-stable distributions are estimated by maximum likelihood on the full sample. Model comparison is conducted using AIC (Akaike Information Criterion) and BIC (Bayesian Information Criterion):

$$AIC = -2\hat{\ell} + 2k$$
$$BIC = -2\hat{\ell} + k \ln n$$

$\hat{\ell}$ = maximum likelihood k = number of free parameters

Both criteria trade off goodness of fit against model complexity, with lower values indicating a preferred specification. Relative to AIC, BIC imposes a stronger penalty on additional parameters.

Table 3.3: Maximum likelihood estimation: comparison of distributions

Distribution	**Parameters**	**Log-Likelihood**	**AIC**	**BIC**	**ΔAIC**
Normal	2	$26{,}992.4$	$-53{,}980.8$	$-53{,}966.7$	$2{,}264.0$
Student-t ($\nu = 2.88$)	3	$28{,}125.4$	$-56{,}244.9$	$-56{,}223.6$	0.0
Stable ($\alpha = 1.57$)	4	$28{,}092.7$	$-56{,}177.4$	$-56{,}149.0$	67.5

Table 3.3 shows that the Student-t distribution provides the best fit, with $\hat{\nu} = 2.88$ degrees of freedom. Relative to the Normal, the improvement is large (ΔAIC $= 2{,}264$), strongly favouring the heavy-tailed specification.

Table 3.4: Maximum likelihood estimates for the Lévy-stable distribution

Parameter	**Estimate**	**Interpretation**
α	1.5692	Stability index (2 = normal, < 2 = heavy tails)
β	-0.1323	Skewness parameter (0 = symmetric)
μ	0.000318	Location parameter
σ	0.005611	Scale parameter

Table 3.4 reports maximum likelihood estimates for the Lévy-stable distribution. The estimated stability parameter indicates substantial tail thickness relative to the Gaussian benchmark. Nevertheless, despite its flexibility in the tails, the Lévy-stable specification does not outperform the Student-t according to the information criteria, which favour the latter as a better balance between fit and parsimony for this sample.

For the crisis-excluded sample, the Student-t estimate increases to $\hat{\nu} = 3.55$, which remains below the threshold of 4 required for finite kurtosis.

Figure 3.3 overlays fitted probability density functions on the empirical histogram of S&P 500 daily returns. The Student-t tracks both the central peak and the tails closely, whereas the Normal underestimates each.

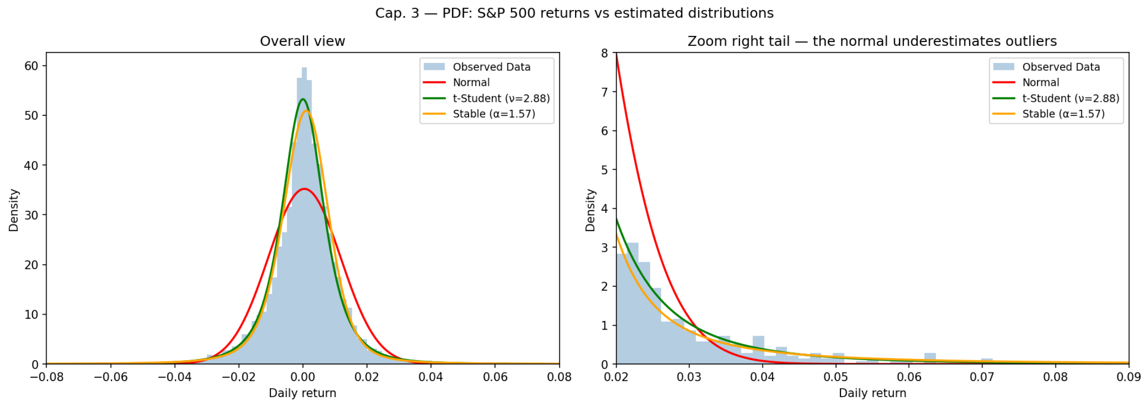

Figure 3.3: Fitted probability density functions overlaid on the empirical histogram of S&P 500 daily returns. The Student-t (green) closely tracks both the central peak and the tails, while the Normal (red) underestimates both.

Figure 3.4 presents QQ plots of S&P 500 daily returns against the Normal, Student-t, and Stable distributions. Deviations from the diagonal indicate departures from the theoretical distribution, particularly in the tails. The Normal distribution shows a pronounced S-shaped deviation in the extremes, confirming its inability to capture tail behaviour. By contrast, the Student-t QQ plot lies substantially closer to the diagonal, consistent with its superior AIC performance.

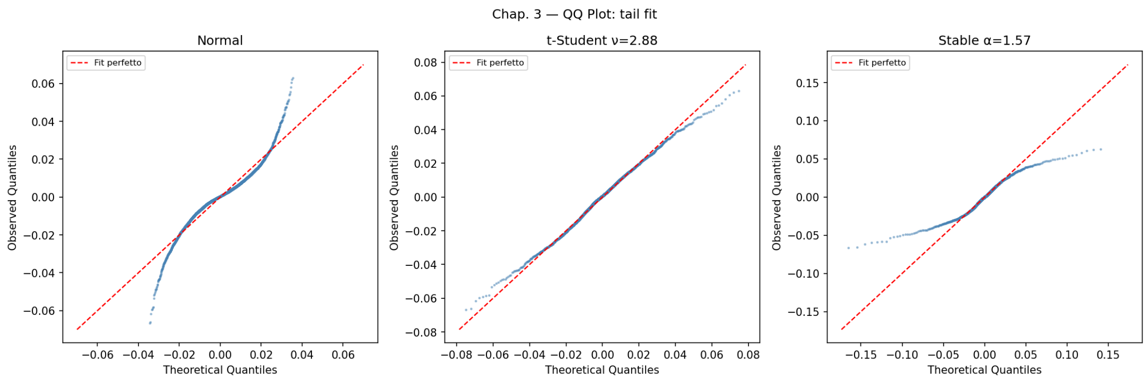

Figure 3.4: QQ plots of S&P 500 daily returns against the Normal (left), Student-t (centre) and Stable (right) distributions. Deviation from the diagonal indicates departure from the theoretical distribution, particularly in the tails.

Chapter 4

Volatility Dynamics: GARCH Modeling

Before estimating GARCH models, volatility clustering is documented formally. Figure 3.1 already suggests that large absolute returns are not uniformly distributed over time but are concentrated around major stress episodes, including the dot-com crash, the Global Financial Crisis, and the COVID crash. Table 4.1 provides formal evidence consistent with this visual pattern. The autocorrelation function of squared returns r_t^2 is positive and statistically significant up to lag 20 (the first five lags are reported), and the ARCH–LM test ($q = 10$) rejects the null of no ARCH effects ($p < 0.001$), supporting the use of a GARCH specification.

Table 4.1: Evidence of volatility clustering

Statistic	**Value**
Autocorrelation of r^2 (lag 1)	0.3157
Autocorrelation of r^2 (lag 2)	0.4205
Autocorrelation of r^2 (lag 3)	0.2443
Autocorrelation of r^2 (lag 4)	0.2951
Autocorrelation of r^2 (lag 5)	0.3135
ARCH–LM statistic ($q = 10$)	2396.18
p-value	0.0000

4.1 The GARCH(1,1) Model

The GARCH(1,1) model of Bollerslev (1986) is estimated under two distributional assumptions for the standardised innovations: a standard Normal distribution ($z_t \sim N(0, 1)$) and a Student-t distribution ($z_t \sim t(\nu)$). Both specifications are estimated by maximum likelihood, implemented in Python via numerical optimisation of the log-likelihood function. The conditional mean is specified as a constant, $r_t = \mu + \varepsilon_t$, with $\varepsilon_t = \sigma_t z_t$.

4.2 Results

Table 4.2 summarises the estimation results. Both specifications imply $\hat{\alpha} + \hat{\beta} = 0.96$, indicating a highly persistent volatility process. The relatively small value of $\hat{\alpha}$ suggests a limited immediate response to new shocks, whereas the large $\hat{\beta}$ implies slow decay of volatility shocks. The implied half-life is approximately 17 trading days, so the impact of a shock on conditional variance requires more than three weeks to halve. The GARCH-t specification improves fit relative to the Gaussian model (ΔAIC $= 464$), indicating that heavy-tailed innovations remain relevant even after allowing for time-varying volatility.

Table 4.2: GARCH(1,1) estimation results

Model	$\hat{\alpha}$	$\hat{\beta}$	$\hat{\alpha} + \hat{\beta}$	$\hat{\nu}$	Log-lik.	AIC	ΔAIC
GARCH-Normal	0.08	0.88	0.96	—	$28{,}871.6$	$-57{,}735.3$	464.3
GARCH-t	0.08	0.88	0.96	5.00	$29{,}104.8$	$-58{,}199.5$	0.0

Figure 4.1 reports the estimated conditional volatility and the corresponding dynamic Value-at-Risk from the GARCH-t model. The top panel plots daily S&P 500 returns together with conditional volatility bands, $\pm 2\hat{\sigma}_t$. The widening of the bands during crisis windows reflects volatility clustering: periods of large price movements tend to be followed by further periods of elevated volatility. This pattern is particularly pronounced during the Global Financial Crisis and the COVID crash.

The middle panel shows annualised conditional volatility. Volatility fluctuates around a historical average of approximately 13.1% in tranquil periods, but rises sharply during stress episodes. Annualised volatility reaches nearly 75% during the 2008 crisis and exceeds 85% during the COVID crash in 2020, confirming that volatility is strongly time-varying and persistent.

The bottom panel compares the dynamic 99% VaR implied by the Gaussian and Student-t GARCH specifications. The GARCH-t model produces wider VaR bands, especially in turbulent periods, reflecting its heavier-tailed innovation distribution. As a result, violation rates are closer to the 1% target than under the Gaussian specification (Table 4.3).

Figure 4.1: GARCH(1,1) results for S&P 500 daily returns 1990–2024.

Table 4.3: Dynamic VaR(99%) performance: GARCH-Normal vs GARCH-t

Model	Average VaR (daily %)	Violation rate
Target level	—	1.00% (88 days)
GARCH-Normal	-2.1587%	2.27% (200 days)
GARCH-t	-2.6964%	1.10% (97 days)

4.3 Standardised Residuals

The distribution of the standardised residuals $\hat{z}_t = \hat{\varepsilon}_t/\hat{\sigma}_t$ provides a diagnostic for the adequacy of the GARCH specification. If time-varying volatility fully accounted for the non-normality in returns, the residuals would be approximately standard normal. Table 4.4 shows that this is not the case.

Under the GARCH specifications, residual kurtosis falls from 13.28 to approximately 4.65–5.15. Volatility clustering therefore accounts for an important share of the non-normality observed in raw returns, but kurtosis remains well above the Gaussian benchmark of 3, indicating that heavy tails persist after controlling for time-varying volatility.

Jarque–Bera tests continue to reject normality for both sets of residuals, with p-values effectively equal to zero. Moreover, the higher residual kurtosis under the GARCH-t specification reflects the fact that the model accommodates heavy-tailed innovations rather than imposing Gaussian behaviour.

Finally, the autocorrelation of squared standardised residuals is close to zero in both cases, indicating that the GARCH model captures the serial dependence in volatility. Even after this dependence is removed, substantial tail risk remains, consistent with the broader evidence developed throughout the thesis.

Table 4.4: Diagnostics on standardised residuals

Statistic	GARCH-Normal	GARCH-t	Expected under $N(0,1)$
Pearson kurtosis	4.6493	5.1523	3.000
Jarque–Bera statistic	1268.6	2099.6	~ 0
Jarque–Bera p-value	0.0000	0.0000	> 0.05
Autocorr. of z_t^2 (lag 1)	0.0019	-0.0117	≈ 0

Chapter 5

Market Evidence: The Implied Volatility Surface

5.1 Data and Methodology

Implied volatility data for S&P 500 index options (SPX) on 21 March 2025 are obtained from OptionMetrics Ivy DB via WRDS. The implied volatility surface is constructed by interpolating across strikes using a smoothed spline fitted to bid–ask midpoints. OptionMetrics reports implied volatilities at standardised delta values

$$\delta \in \{-90, -85, \ldots, -10, 10, \ldots, 85, 90\}$$

for maturities of 30, 60, 91, and 182 calendar days. The date 21 March 2025 corresponds to the standard monthly expiry Friday, when option market liquidity is typically high.

Delta is used as the moneyness axis rather than the strike, as it provides a standardised, scale-free measure of distance from at-the-money. In particular, $\delta = -50$ (put) and $\delta = +50$ (call) correspond to at-the-money options, while $\delta = -10$ (put) corresponds to deep out-of-the-money protection against severe market declines.

5.2 The Volatility Skew

Figure 5.1 reports implied volatility smiles across the four maturities. Under Black–Scholes, implied volatility should be constant across strikes for a given maturity; empirically, the surface displays a pronounced asymmetry. Put options with $\delta = -25$ trade at implied volatilities of 19–20%, compared with 13–15% for call options with $\delta = +25$, with an at-the-money level around 16%. This asymmetry is consistent with the negative skewness documented in historical returns.

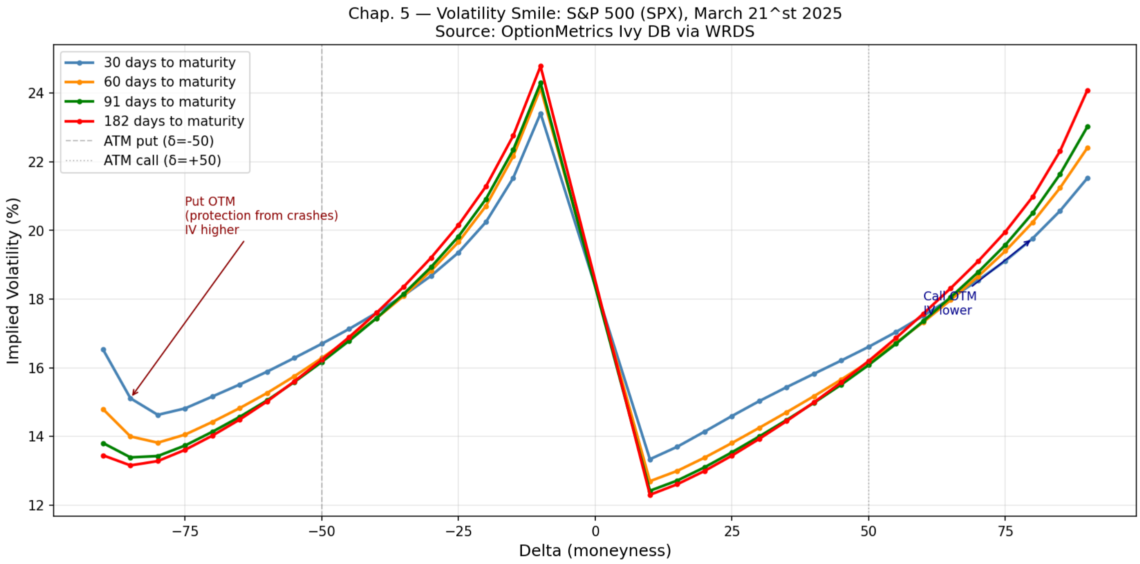

Figure 5.1: Implied volatility smile for S&P 500 options (SPX), 21 March 2025. Horizontal axis: option delta (negative = put, positive = call). Vertical axis: implied volatility (%). Source: OptionMetrics Ivy DB via WRDS.

Figure 5.2 focuses on the put side by zooming in on the left tail. The figure highlights the fat-tail premium, i.e., the excess implied volatility above the ATM benchmark that the market assigns to options that insure against large downside moves.

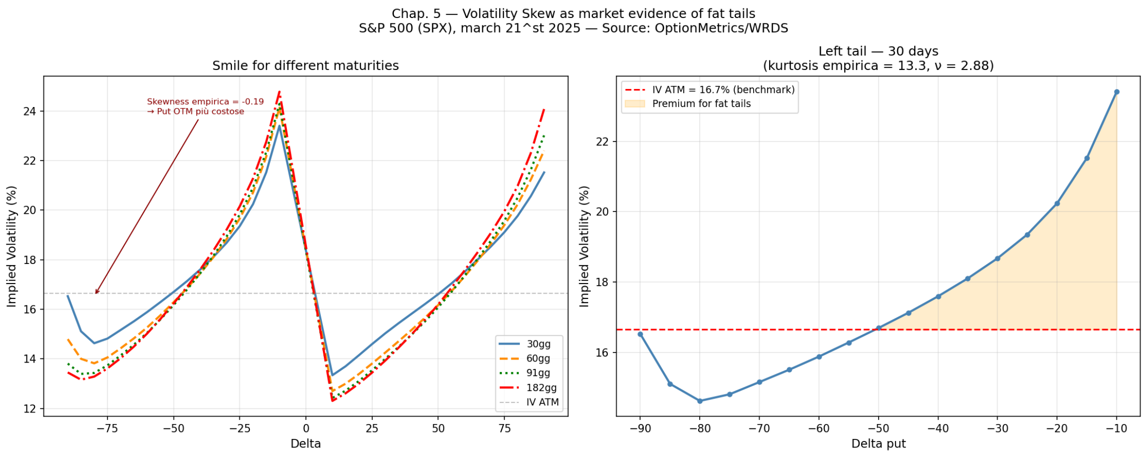

Figure 5.2: Left panel: volatility smile across four maturities with skewness annotation. Right panel: put-side implied volatility for 30-day options, with the orange area representing the fat-tail premium above the ATM benchmark of 16.7%.

5.3 Quantitative Connection to Historical Return Evidence

Table 5.1 reports the risk reversal and butterfly spread across maturities. Three features are particularly relevant.

First, the risk reversal is negative at all maturities, ranging from -4.76% at 30 days to -6.71% at 182 days. This pattern is consistent with the negative skewness of -0.187 documented in historical returns (Chapter 3) and reflects the market's willingness to pay a premium for crash protection.

Second, the butterfly spread is positive and increases with maturity, from $+0.32\%$ to $+0.58\%$. Deep out-of-the-money options therefore trade at implied volatilities above those predicted by a lognormal benchmark, suggesting that the market assigns greater probability to extreme outcomes than under Gaussian assumptions. This is consistent with the excess kurtosis of 13.28 in raw returns and 5.15 in GARCH standardised residuals (Chapter 4).

Third, put options with $\delta = -10$ trade at implied volatilities of 23–25%, implying a premium of roughly 40% relative to ATM volatility. The persistence of this premium across maturities indicates that investors treat heavy tails as a structural feature of the return-generating process rather than a transitory crisis-related distortion.

The maturity pattern of the skew is also informative. If fat tails reflected short-lived crisis dynamics, the skew would be expected to flatten at longer maturities. Instead, downside protection becomes relatively more expensive over longer horizons, consistent with the historical return evidence showing that heavy tails remain significant even after excluding crisis periods and time-varying volatility.

Overall, the options market provides independent, forward-looking confirmation that tail risk is not a temporary by-product of volatility clustering alone but a persistent structural feature of equity returns.

Table 5.1: Implied volatility surface: risk reversal and butterfly across maturities

Maturity	IV ATM	IV put 25Δ	IV call 25Δ	IV put 10Δ	RR	BF
30 days	16.65%	19.36%	14.60%	23.41%	-4.76%	$+0.32\%$
60 days	16.23%	19.66%	13.81%	24.14%	-5.85%	$+0.51\%$
91 days	16.13%	19.82%	13.54%	24.29%	-6.29%	$+0.56\%$
182 days	16.22%	20.16%	13.44%	24.78%	-6.71%	$+0.58\%$

Chapter 6

Conclusions

This thesis presents a comprehensive empirical assessment of the normality hypothesis for S&P 500 daily returns. Across the different methodologies considered and under alternative sample definitions, the evidence against normality remains strong.

Three formal tests—Jarque–Bera, Anderson–Darling, and chi-square goodness-of-fit—reject normality at conventional significance levels in both the full sample ($n = 8{,}817$) and the crisis-excluded sample ($n = 7{,}938$), with p-values effectively equal to zero. The empirical kurtosis declines from 13.28 to 6.46 when crisis windows are removed, but it remains more than twice the Gaussian benchmark of 3. This indicates that heavy tails are a structural feature of the return-generating process rather than an artefact of a small number of disruptive episodes.

Among the distributions estimated by maximum likelihood, the Student-t provides the best fit, with $\hat{\nu} = 2.88$ degrees of freedom and an improvement over the Normal of ΔAIC $= 2{,}264$. The estimate implies theoretically infinite kurtosis, consistent with the empirical instability of higher moments across subsamples. The GARCH(1,1) results further show that accounting for time-varying volatility reduces, but does not eliminate, non-normality: residual kurtosis falls from 13.28 to 5.15 under GARCH-t, yet fat tails persist in standardised residuals. In risk-management terms, the consequences are material. The GARCH-Normal model produces a VaR(99%) violation rate of 2.27%, more than double the target, whereas the GARCH-t achieves 1.10%.

The implied volatility surface of SPX options on 21 March 2025 provides complementary, market-based evidence. The 25-delta risk reversal ranges from -4.76% to -6.71% across maturities, and the butterfly spread is positive at all tenors, consistent with the negative skewness and excess kurtosis documented in historical returns. The monotonic steepening of the skew with maturity further indicates that markets price tail risk as a persistent, structural feature.

Overall, evidence from three sources—historical return distributions, GARCH modelling of conditional volatility, and the implied volatility surface—points to the same conclusion: equity returns exhibit heavy tails that are structural, persistent, and priced in the market. As a result, the Gaussian assumption is inadequate for both statistical modelling and practical risk management.

Limitations. Two limitations merit acknowledgment. The implied volatility analysis is based on a single trading date and may not be representative of other market conditions. In addition, the GARCH(1,1) specification, while standard, does not capture asymmetric (leverage) effects or long memory in volatility.

Future research. Natural extensions include estimating asymmetric GARCH specifications (GJR-GARCH, EGARCH), incorporating realised volatility measures, and extending the implied-volatility analysis to a panel of dates in order to test whether option-implied non-normality co-moves with tail-risk measures from historical returns.

More broadly, the evidence in this thesis suggests that heavy tails should not be treated as anomalous deviations from normality but as a fundamental characteristic of financial markets.

Appendix A

Implementation Details

All empirical analyses were implemented in Python using NumPy, pandas, SciPy, matplotlib, and custom maximum likelihood optimisation routines. Statistical tests, distribution fitting, GARCH estimation, VaR backtesting, and implied volatility calculations were performed programmatically on the full dataset. All tables and figures reported throughout the thesis were generated directly from the Python output to ensure reproducibility and consistency of results.

The estimation of the Lévy-stable distribution required numerical maximum likelihood optimisation due to the absence of a closed-form density, resulting in significantly higher computational complexity relative to the Gaussian and Student-t specifications.

A.1 Use of Artificial Intelligence Tools

Generative AI tools were used exclusively for language editing, stylistic refinement, and formatting support. All research design, data collection, empirical analyses, coding, interpretation of results, and conclusions were developed independently by the author.

Bibliography

Anderson, T. W. and D. A. Darling (June 1952). "Asymptotic Theory of Certain "Goodness of Fit" Criteria Based on Stochastic Processes". In: *The Annals of Mathematical Statistics* 23.2, pp. 193–212. ISSN: 0003-4851. (Visited on 05/18/2026).

Bachelier, L (1900). "Théorie de La Spéculation". In: *Annales Scientifiques de l'École Normale Supérieure* 17, pp. 21–86.

Black, Fischer and Myron Scholes (1973). "The Pricing of Options and Corporate Liabilities". In: *Journal of Political Economy* 81, pp. 637–654.

Bollerslev, Tim (Aug. 1987). "A Conditionally Heteroskedastic Time Series Model for Speculative Prices and Rates of Return". In: *The Review of Economics and Statistics* 69.3, p. 542. ISSN: 00346535. JSTOR: 1925546. (Visited on 05/18/2026).

Cont, Rama (2001). "Empirical Properties of Asset Returns: Stylized Facts and Statistical Issues". In: *Quantitative Finance* 1, pp. 223–236.

Engle, Robert F. (1982). "Autoregressive Conditional Heteroscedasticity with Estimates of the Variance of United Kingdom Inflation". In: *Econometrica* 50, pp. 987–1007.

Fama, Eugene F. (Jan. 1965). "The Behavior of Stock-Market Prices". In: *The Journal of Business* 38.1, p. 34. ISSN: 0021-9398, 1537-5374. JSTOR: 2350752. (Visited on 05/18/2026).

Gabaix, Xavier (Sept. 2009). "Power Laws in Economics and Finance". In: *Annual Review of Economics* 1.1, pp. 255–294. ISSN: 1941-1383, 1941-1391. (Visited on 05/18/2026).

Hamao, Yasushi, Ronald W. Masulis, and Victor Ng (Apr. 1990). "Correlations in Price Changes and Volatility across International Stock Markets". In: *Review of Financial Studies* 3.2, pp. 281–307. ISSN: 0893-9454, 1465-7368. (Visited on 05/18/2026).

Jarque, Carlos M. and Anil K. Bera (Aug. 1987). "A Test for Normality of Observations and Regression Residuals". In: *International Statistical Review / Revue Internationale de Statistique* 55.2, p. 163. ISSN: 03067734. JSTOR: 1403192. (Visited on 05/18/2026).

Mandelbrot, Benoit (1963). "The Variation of Certain Speculative Prices". In: *The Journal of Business* 36.4, pp. 394–419. JSTOR: 2350970.

— (2004). *The Misbehavior of Markets*.

Markowitz, Harry (1952). “Portfolio Selection”. In: *The Journal of Finance* 7.1, pp. 77–91.

Praetz, Peter D. (1972). “The Distribution of Share Price Changes”. In: *The Journal of Business* 45.1, pp. 49–55.